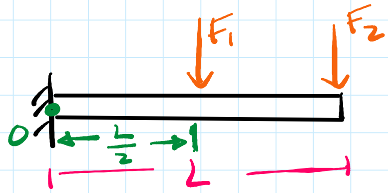
4. Force System Resultants

Tuesday, September 3, 2024 9:39 AM

* Shift from a particle analysis to a ~~rigid~~ body analysis (bodies that do not deform under loading)

* Leverage
* Application of force matters (position of force on body)

* Bending Moment → (Torque) N·m, lb·ft



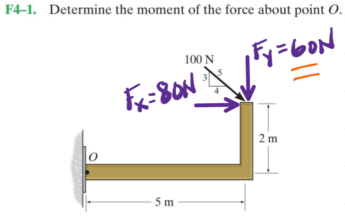
* Bending moment is an internal force that the beam experiences
* Resistance to bending

$F_1 = F_2$

* Which force will produce a bigger bending moment on the beam about point O?

$$\circlearrowleft + \; M_O = -L \cdot F_2 - \frac{L}{2} \cdot F_1$$

F4–1. Determine the moment of the force about point *O*.



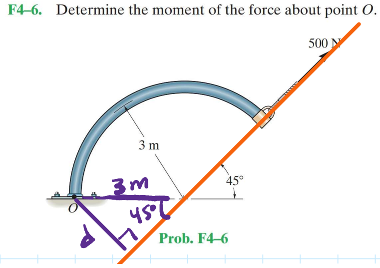
Prob. F4–1

$$\circlearrowleft + \; M_O = -80(2) - 60(5)$$

$$= -160 - 300$$

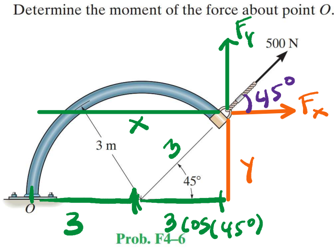
$$M_O = -460 \text{ N·m}$$

F4–6. Determine the moment of the force about point *O*.

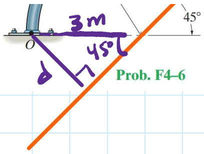


Prob. F4–6

F4–6. Determine the moment of the force about point *O*.



Prob. F4–6



Prob. F4–6

Prob. F4–6

$M_O = d \cdot F$

$d = 3\sin(45°)$
$d = 2.12\ m$

$M_O = (2.12)(500)$

$M_O = 1060.66\ N\cdot m$

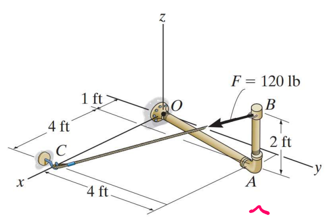
$M_O = x F_y - y F_x$

$\left.\begin{array}{l} x = 3 + 3\cos(45°) \\ F_y = 500\sin(45°) \end{array}\right\}$ $1810.66\ N\cdot m$

$\left.\begin{array}{l} y = 3\sin(45°) \\ F_x = 500\cos(45°) \end{array}\right\}$ $750\ N\cdot m$

$M_O = 1810.66 - 750 = 1060.66\ N\cdot m$

F4–11. Determine the moment of force **F** about point *O*. Express the result as a Cartesian vector.



Prob. F4–11

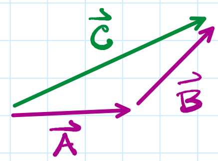
$\hat{u}_{BC}$

$\vec{M} = \underline{\underline{\vec{r}}} \times \vec{F} \rightarrow$ Force vector

position vector $\rightarrow \vec{r}_{OB}$

$O \rightarrow B$

Add vectors head to tail

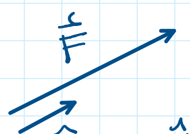


$\vec{C} = \vec{A} + \vec{B}$

$\langle x, y, z\rangle$ $\langle x, y, z\rangle$

$\vec{M}_O = \vec{r}_{OB} \times \vec{F}$

"easy"



$\hat{u} \rightarrow \hat{u}$ has a magnitude of 1.
$\langle x, y, z\rangle$

$\vec{F} = \underline{\underline{|F|}} \cdot \hat{u}$

mag. of F

unit vector that captures direction on $\vec{F}$.

$\vec{r}_{OB} = 1\hat{i} + 4\hat{j} + 2\hat{k}$

$\vec{r}_{OC} = 5\hat{i} + 0\hat{j} + 0\hat{k}$

$\vec{r}_{OB} + \vec{r}_{BC} = \vec{r}_{OC}$

$\vec{r}_{BC} = \vec{r}_{OC} - \vec{r}_{OB}$

$\vec{r}_{BC} = 5\hat{i} - (1\hat{i} + 4\hat{j} + 2\hat{k})$

$\vec{r}_{BC} = 4\hat{i} - 4\hat{j} - 2\hat{k}$

ft ft ft

$\hat{u}_{BC} = \dfrac{\vec{r}_{BC}}{\|r_{BC}\|} = \dfrac{4\hat{i} - 4\hat{j} - 2\hat{k}}{\sqrt{4^2 + (-4)^2 + (-2)^2}} = \frac{4}{6}\hat{i} - \frac{4}{6}\hat{j} - \frac{2}{6}\hat{k}$

ft

$\hat{u}_{BC} = \frac{2}{3}\hat{i} - \frac{2}{3}\hat{j} - \frac{1}{3}\hat{k}$

$\vec{F} = F \cdot \hat{u}_{BC}$

$$\underbrace{u_{BC} = \tfrac{2}{3}\hat{i} - \tfrac{2}{3}\hat{j} - \tfrac{1}{3}\hat{k}}$$ $\quad F = F \cdot u_{BC}$

$$\vec{F} = \underset{\text{lbs}}{120}\left(\underbrace{\tfrac{2}{3}\hat{i} - \tfrac{2}{3}\hat{j} - \tfrac{1}{3}\hat{k}}_{\text{unit less}}\right) = 80\hat{i} - 80\hat{j} - 40\hat{k} \text{ lbs}$$

$$\vec{M}_O = \vec{r}_{OB} \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 4 & 2 \\ 80 & -80 & -40 \end{vmatrix} = \begin{array}{l} (4(-40) - 2(-80))\hat{i} \\ -(1(-40) - (2)(80))\hat{j} \\ +((1)(-80) - 4(80))\hat{k} \end{array}$$

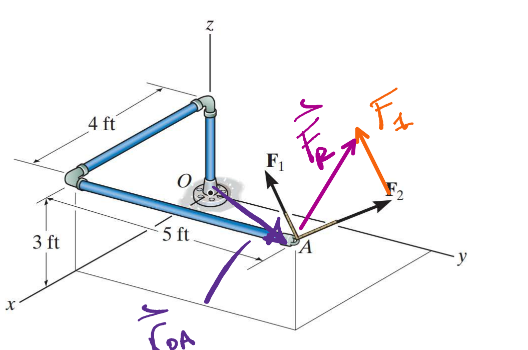
$$\vec{M}_O = (-160+160)\hat{i} - (-40-160)\hat{j} + (-80-320)\hat{k}$$

$$\boxed{\vec{M}_O = +200\hat{j} - 400\hat{k} \text{ lbs}\cdot\text{ft}}$$

c.c.w about y-axis

c.w. about z-axis

F4–12. If the two forces $\mathbf{F}_1 = \{100\mathbf{i} - 120\mathbf{j} + 75\mathbf{k}\}$ lb and $\mathbf{F}_2 = \{-200\mathbf{i} + 250\mathbf{j} + 100\mathbf{k}\}$ lb act at A, determine the resultant moment produced by these forces about point O. Express the result as a Cartesian vector.



Prob. F4–12

$$\vec{M}_O = \vec{r}_{OA} \times \vec{F}_1 + \vec{r}_{OA} \times \vec{F}_2$$

$$\vec{M}_O = \vec{r}_{OA} \times (\vec{F}_1 + \vec{F}_2)$$

$$\vec{F}_R = \vec{F}_1 + \vec{F}_2 = (-100\hat{i} + 130\hat{j} + 175\hat{k}) \text{ lbs}$$

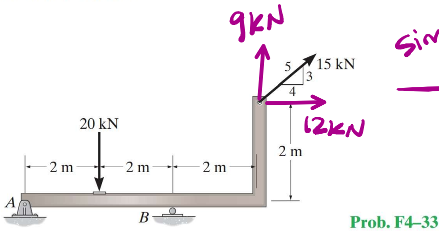
$$\vec{r}_{OA} = (4\hat{i} + 5\hat{j} + 3\hat{k}) \cdot \text{ft}$$

$$\vec{M}_O = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & 5 & 3 \\ -100 & 130 & 175 \end{vmatrix} = \begin{array}{l} (5(175) - 3(130))\hat{i} \\ -(4(175) - 3(-100))\hat{j} \\ +(4(130) - 5(-100))\hat{k} \end{array}$$

$$(875-390)\hat{i} - (700+300)\hat{j} + (520+500)\hat{k}$$

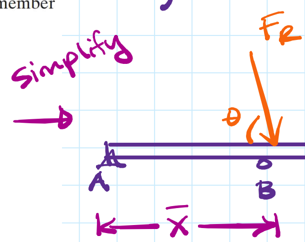
$$\boxed{485\hat{i} - 1000\hat{j} + 1020\hat{k} \text{ lbs}\cdot\text{ft}}$$

F4–33. Replace the loading system by an equivalent resultant force and specify where the resultant's line of action intersects the horizontal segment of the member measured from A.

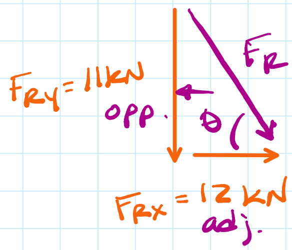


Prob. F4–33

Objective



$$\sqrt{F_{Rx}^2 + F_{Ry}^2} = F_R$$



$F_{Ry} = -20kN + 9\ kN$

$F_{Ry} = -11\ kN = 11\ kN \downarrow$

$F_{Rx} = 12\ kN \rightarrow$

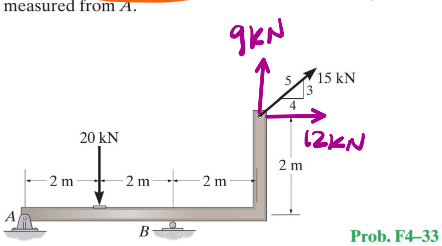
$F_R = \sqrt{(-11)^2 + (12)^2} = \sqrt{121 + 144}$

$F_R = 16.3\ kN$ ← magnitude

$\tan\theta = \frac{11}{12} \rightarrow \theta = \tan^{-1}\left(\frac{11}{12}\right)$

$\theta = 42.5°$ ← direction

F4–33. Replace the loading system by an equivalent resultant force and specify where the resultant's line of action intersects the horizontal segment of the member measured from A.

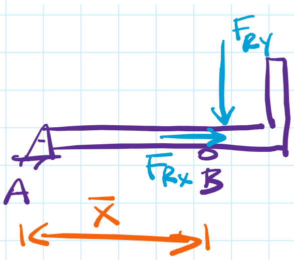


Prob. F4–33

$M_A = -(2)(20) - 2(12) + (6)(9)$

$M_A = -40 - 24 + 54$

$$\boxed{M_A = -10\ kN \cdot m = 10kN \cdot m \circlearrowright}$$



$F_R = 16.3kN \quad \theta = 42.5°$

$M_A = -\bar{x} \cdot F_{Ry} = -10\ kN \cdot m$

$\bar{x} = \frac{10}{F_{Ry}} = \frac{10}{11}\ kN \cdot m$

$\bar{x} = 0.909\ m$

* Distributed Loading

F4–42. Determine the resultant force and specify where it acts on the beam measured from A.

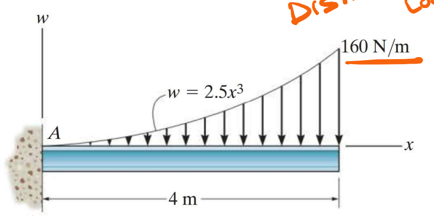
Distributed loading

$$F_R = \int_L w(x)\, dx$$

$\frac{N}{\cancel{m}} \cdot \cancel{m}$

F4–42. Determine the resultant force and specify where it acts on the beam measured from A.

Distributed Loading



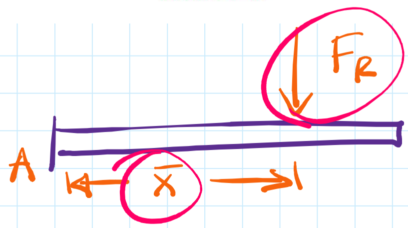
Prob. F4–42

$$F_R = \int_L w(x)\,dx$$

$$\bar{x} = \frac{\int_L x \cdot \boxed{w(x)\,dx}}{F_R}$$

[m] [N]

equilibrium analysis



$$\bar{x} \cdot F_R = \int \underbrace{x}_{\text{distance}} \cdot \underbrace{w\,dx}_{\text{Force}}$$

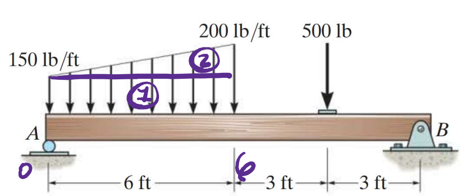
$$F_R = \int_0^4 2.5x^3\,dx = \frac{2.5}{4} x^4 \Big|_0^4 = \frac{2.5}{\cancel{4}} (4)^{\cancel{4}\,3} - \cancel{\frac{2.5}{4}(0)}^{\,0}$$

$$F_R = \frac{5}{2}(4^3) = \frac{5}{\cancel{2}}(\cancel{64}\,32) = 160\text{ N}$$

$$\bar{x} = \frac{\int_L x \cdot w(x)\,dx}{F_R} = \frac{\int_0^4 x \cdot 2.5x^3\,dx}{160} = \frac{\int_0^4 2.5x^4\,dx}{160}$$

$$\bar{x} = \frac{1}{160}\left(\frac{2.5}{5} x^5 \Big|_0^4\right) = \frac{1}{160}\left(\frac{2.5}{5}(4)^5\right) = 3.2\text{ m}$$

F4–40. Determine the resultant force and specify where it acts on the beam measured from A.



Prob. F4–40

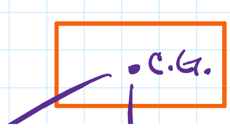
Centroid (Center of mass/gravity)

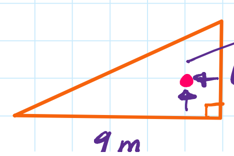
→ geometric center

→ same as center of mass for homogeneous bodies,

→ same as center of gravity for constant gravitational fields

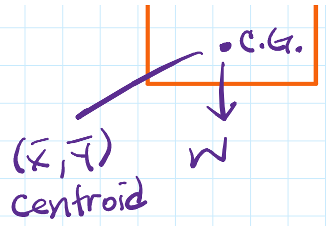
$w(x) = mx + b$

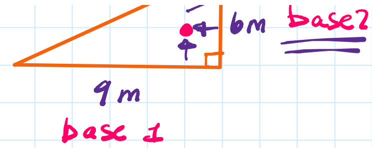


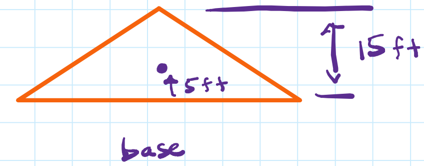


$$W(x) = m x + \underline{\underline{b}}$$
Y

$$m = \frac{\text{rise}}{\text{run}} = \frac{50}{6} = \frac{25}{3}$$



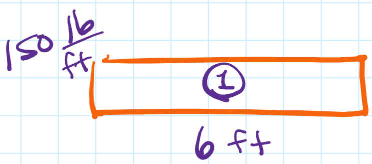


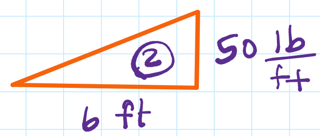


$$150 = \frac{25}{3}(0) + b \rightarrow b = 150$$

$$W(x) = \frac{25}{3}x + 150$$

→ integrate to solve for F_R



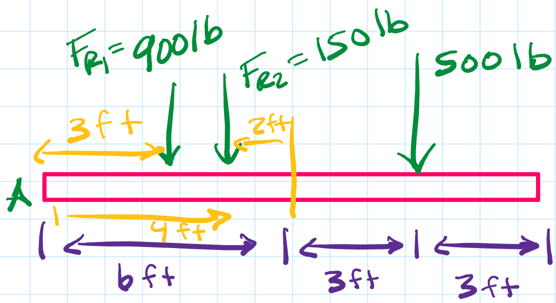


$$F_{R1} = 150\frac{\text{lb}}{\text{ft}}(6\text{ ft})$$

$$F_{R1} = 900 \text{ lb}$$

$$F_{R2} = \frac{1}{2}(6\text{ft})(50\text{ft})$$

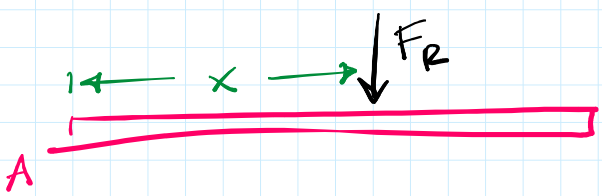
$$F_{R2} = 150 \text{ lb}$$



$$F_R = 1550 \text{ lb}$$

$$M_A = -3(900) - 4(150) - 9(500)$$

$$M_A = -7800 \text{ lb}\cdot\text{ft}$$



$$\bar{x} \cdot F_R = 7800$$

$$\bar{x} = \frac{7800}{1550} = 5.03 \text{ ft}$$